

EE 446: TinyML

Lab 6: TinyML for Anomaly Detection

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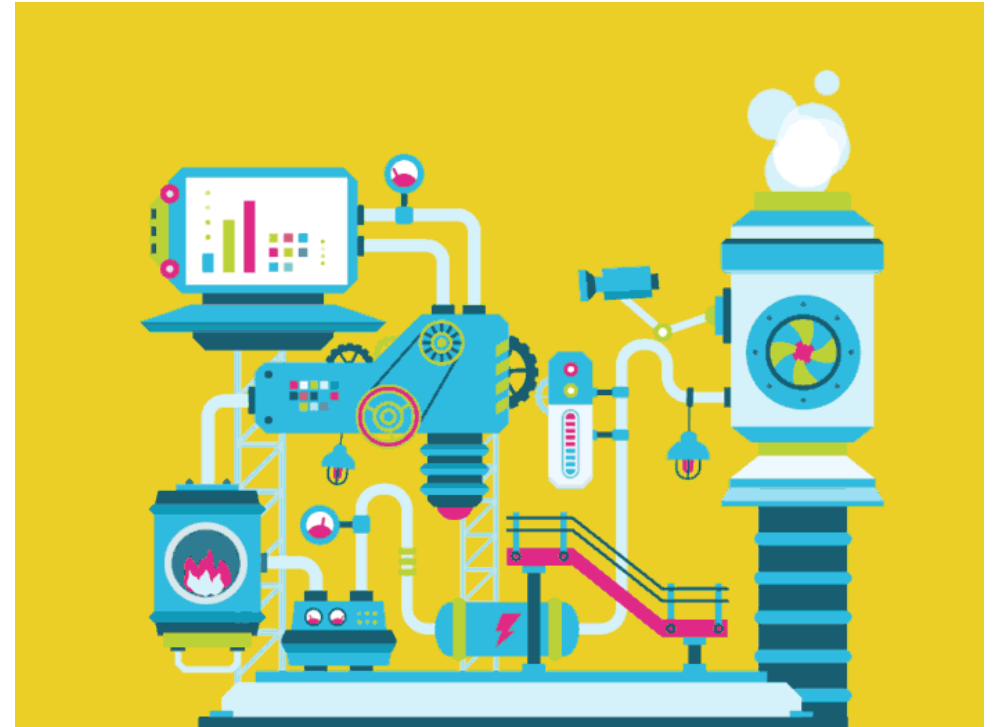
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Lab 6

- **Two Part Lab**
- **Part 1 (Edge Impulse):** Anomaly detection in fans using Arduino Nano BLE 33 Sense - NN Classifier + K-means
- **Part 2 (Tensorflow Lite – Python Notebook):** Anomaly detection in human activities using Arduino Nano BLE 33 Sense – Autoencoder + K-means



Lab 6: Prat 1 - Software - Road Map

1. Clone ***Edge Impulse*** Project
2. Examine and understand ML Pipeline
3. Train a classifier model for fan operating status detection and perform model compression (int 8 quantization)
4. Train an anomaly detector using k-means



Lab 6: Prat 1 – Software Lab – Edge Impulse

- Open Edge Impulse Fan Monitoring example from:

<https://studio.edgeimpulse.com/public/47996/latest>

- Click on 'Clone this Project'.
- Register for an account on Edge Impulse (*If you haven't done yet*).

Lab 6: Prat 1 – Software Lab – Edge Impulse

- On the left pane, Go to 'Create Impulse'.

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Time series data, Spectral Analysis, Classification (Keras), and Anomaly Detection (K-means). Below these is an 'Upgrade Plan' section. The main workspace contains five colored panels: 1. 'Time series data' (red) with settings for input axes (accX, accY, accZ), window size (2,000 ms), window increase (stride) (220 ms), frequency (100 Hz), and zero-pad data (checked). 2. 'Spectral Analysis' (light blue) with a name field 'Spectral features' and input axes (accX, accY, accZ) all checked. 3. 'Classification (Keras)' (purple) with a name field 'NN Classifier', input features (Spectral features) checked, and output features (2 nominal, off). 4. 'Anomaly Detection (K-means)' (purple) with a name field 'Anomaly detection', input features (Spectral features) checked, and output features (1 Anomaly score). 5. 'Output features' (green) showing 3 features (nominal, off, Anomaly score) and a 'Save Impulse' button. A dashed box at the bottom center says 'Add a processing block'.

Explore this carefully and provide answers in your lab report

Lab 6: Prat 1 – Software Lab – Edge Impulse

- On the left pane, Go to ‘Spectral Features’.
- Then click “Generate Features”

The screenshot displays the Edge Impulse web interface. On the left, a sidebar menu lists various options, with 'Spectral features' highlighted. The main area is titled 'Parameters' and 'Generate features'. It includes a 'Training set' section with fields for 'Data in training set' (3m 30s), 'Classes' (2 (nominal, off)), 'Training windows' (350), 'Calculate feature importance' (checked), and 'Normalize features' (set to 'Don't normalize data'). A red box highlights the 'Generate features' button. Below this is the 'Feature generation output' section, which currently shows '(0)'. To the right, the 'Feature explorer' section displays a scatter plot with two clusters of data points, one blue (nominal) and one orange (off). Below the plot is the 'Feature importance' section, which shows 'accX RMS' and a dropdown menu set to 'All data'.

Explore this carefully and provide answers in your lab report

Lab 6: Prat 1 – Software Lab – Edge Impulse

- On the left pane, Go to ‘NN Classifier’.
- Then click “Save & Train”

The screenshot displays the Edge Impulse web interface. On the left, a sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Time series data, Spectral features. Below these is an 'Upgrade Plan' section. The main area is divided into two panels. The left panel, titled 'Neural Network settings', contains 'Training settings' (Number of training cycles: 30, Use learned optimizer: unchecked, Learning rate: 0.0005, Training processor: CPU) and 'Advanced training settings'. Below this is the 'Neural network architecture' section, which shows a sequence of layers: 'Input layer (33 features)', 'Dense layer (32 neurons)', 'Dense layer (16 neurons)', 'Add an extra layer', and 'Output layer (2 classes)'. At the bottom of this panel is a red-bordered button labeled 'Save & train'. The right panel, titled 'Training output', shows the 'Model' version as 'Quantized (int8)'. It displays 'Last training performance (validation set)' with 'ACCURACY 100.0%' and 'LOSS 0.13'. Below this is a 'Confusion matrix (validation set)' table:

	NOMINAL	OFF
NOMINAL	100%	0%
OFF	0%	100%
F1 SCORE	1.00	1.00

Below the confusion matrix is a 'Metrics (validation set)' table:

METRIC	VALUE
Area under ROC Curve	1.00
Weighted average Precision	1.00
Weighted average Recall	1.00
Weighted average F1 score	1.00

At the bottom of the right panel is a 'Data explorer (full training set)' section showing a scatter plot with a legend: nominal - correct (green), off - correct (yellow), and off - incorrect (red).

Explore this carefully and provide answers in your lab report



Lab 6: Prat 1 – Software Lab – Edge Impulse

- On the left pane, Go to ‘Anomaly Detection’.
- Then click “Save & Train”

EDGE IMPULSE

Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Time series data, Spectral features
NN Classifier
Anomaly detection
Retrain model

Upgrade Plan
Get access to higher job limits and more collaborators.
View plans

☒ accX Peak 1 Height ★
☐ accX Peak 2 Freq
☐ accX Peak 2 Height
☐ accX Peak 3 Freq
☐ accX Peak 3 Height
☐ accX Spectral Power 0.1 - 0.5
☐ accX Spectral Power 0.5 - 1.0
☐ accX Spectral Power 1.0 - 2.0
☐ accX Spectral Power 2.0 - 5.0
☐ accY RMS
☐ accY Peak 1 Freq
☐ accY Peak 1 Height
☐ accY Peak 2 Freq
☐ accY Peak 2 Height
☐ accY Peak 3 Freq
☐ accY Peak 3 Height
☐ accY Spectral Power 1.0 - 2.0
☒ accY Spectral Power 2.0 - 5.0 ★
☐ accZ RMS
☐ accZ Peak 1 Freq
☐ accZ Peak 1 Height
☐ accZ Peak 2 Freq
☐ accZ Peak 2 Height
☐ accZ Peak 3 Freq
☐ accZ Peak 3 Height
☐ accZ Spectral Power 0.1 - 0.5
☐ accZ Spectral Power 0.5 - 1.0
☐ accZ Spectral Power 1.0 - 2.0
☒ accZ Spectral Power 2.0 - 5.0 ★

Save & train

Explore this carefully and provide answers in your lab report

Lab 6: Prat 1 – Software Lab – Edge Impulse

- After training is completed, click on 'Deployment' on the left pane.

- Choose Arduino library -> [Arduino Nano 33 BLE Sense](#)

(see next slide for more information)

- Choose [EON Compiler](#) -> Build. *(see next slide for more information)*

- Download the script for flashing on the hardware.

Lab 6: Prat 1 – Model Deployment – Edge Impulse

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Time series data, S...

Create impulse

Spectral features

NN Classifier

Anomaly detection

Retrain model

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Arduino Nano 33 BLE Sense

SELECTED DEPLOYMENT

Arduino Nano 33 BLE Sense

A binary containing both the Edge Impulse data acquisition client and your full impulse.

MODEL OPTIMIZATIONS

Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler

Same accuracy, 36% less RAM, 40% less ROM.

Quantized (int8)

Selected ✓

	SPECTRAL FEATUR...	NN CLASSIFIER	TOTAL
LATENCY	18 ms.	1 ms.	19 ms.
RAM	8.1K	1.7K	8.1K
FLASH	-	20.0K	-
ACCURACY			-

Latest build

v12 (Arduino Nano 33 BLE Sense)

Today, 16:00:06

View docs

Build output

Cancel

Creating job... OK (ID: 32219443)

✓Job scheduled at 29 Apr 2025 23:06:57

✓Job started at 29 Apr 2025 23:07:00

Writing templates...

✓Job scheduled at 29 Apr 2025 23:07:07

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Lab 6: Prat 1 – Model Deployment – Edge Impulse

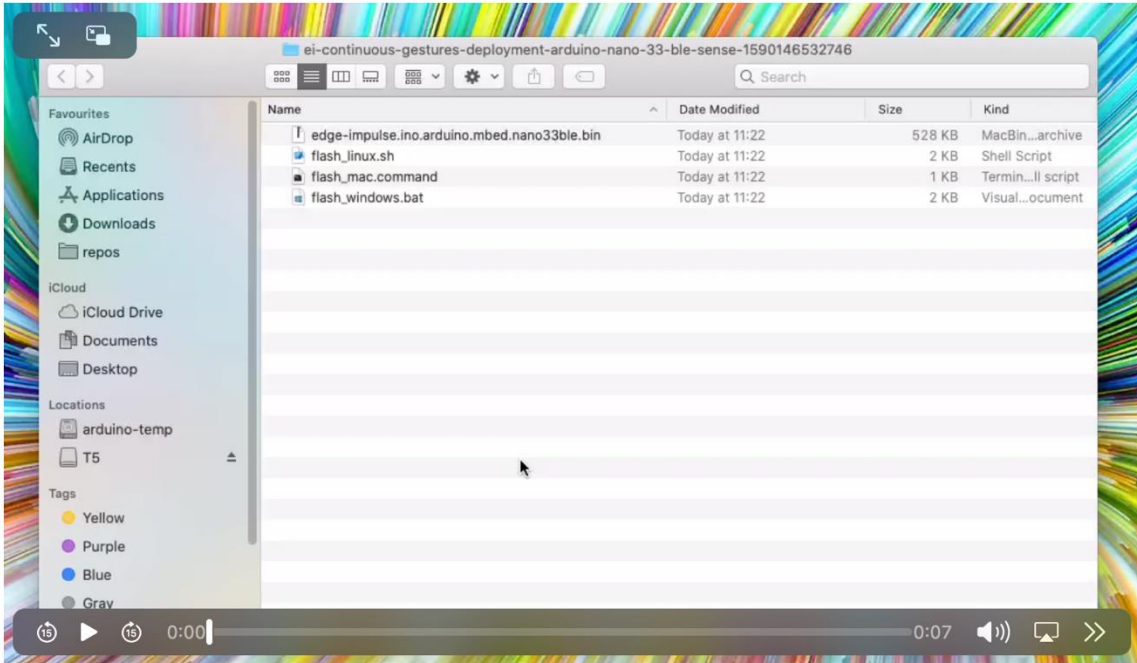
Upon a Successful
Model Build You
Should Be Able to See
Something Like This





Built Arduino Nano 33 BLE Sense firmware

Click on the script for your operating system to flash the binary.



Name	Date Modified	Size	Kind
edge-impulse.ino.arduino.mbed.nano33ble.bin	Today at 11:22	528 KB	MacBin...archive
flash_linux.sh	Today at 11:22	2 KB	Shell Script
flash_mac.command	Today at 11:22	1 KB	Termin...ll script
flash_windows.bat	Today at 11:22	2 KB	Visual...ocument

```
bytes (27%) of program storage
83040 bytes.
e 58624 bytes (22%) of dynamic
520 bytes for local variables.
ytes.

arduino-nano-33-ble-sense done
```

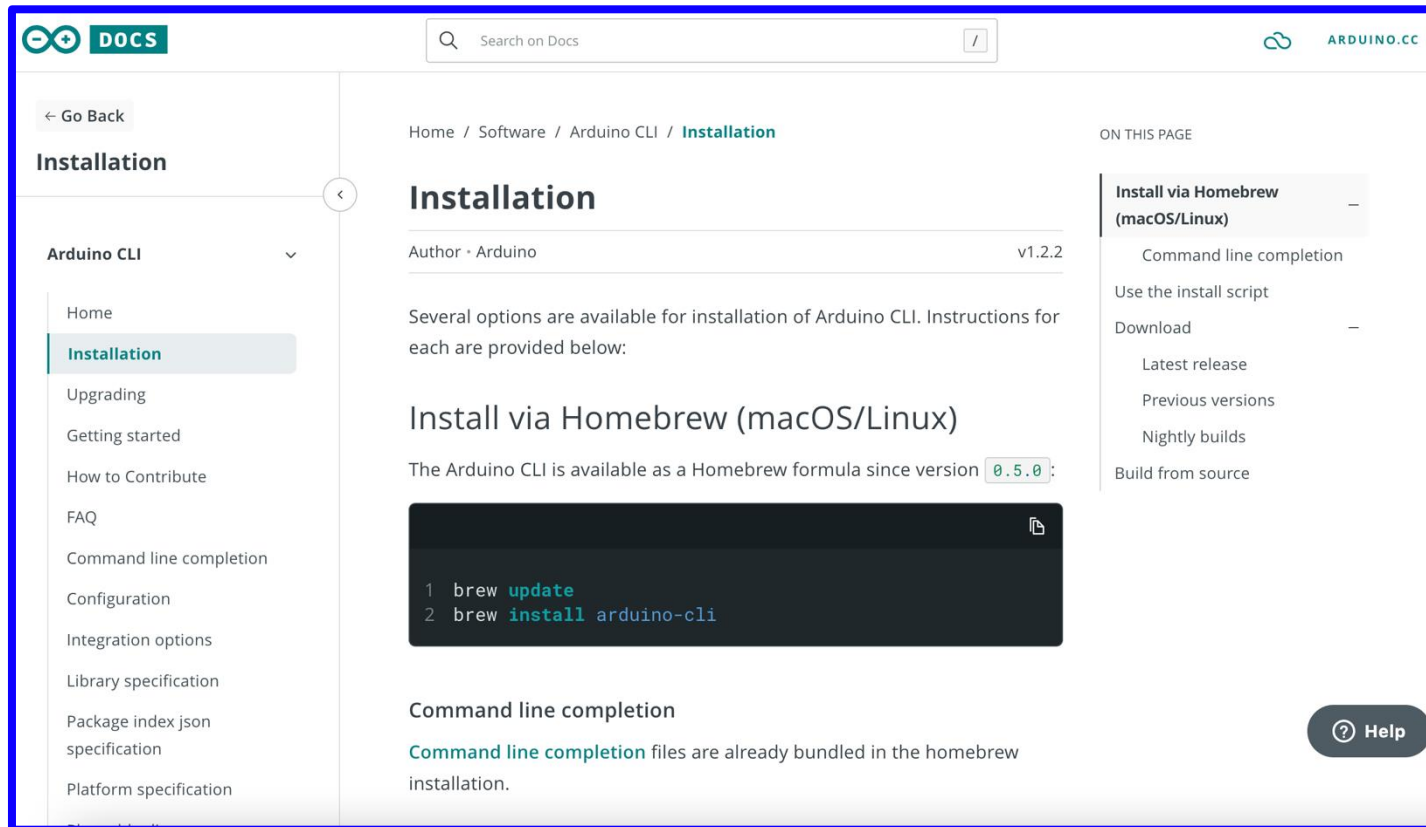
Lab 6: Prat 1 - Hardware - Road Map

1. Connect Arduino Nano BLE to your computer
2. Install Arduino CLI and Edge Impulse CLI
2. Flash fan anomaly detection compressed model obtained from Edge Impulse to Arduino Nano BLE
4. Use Serial monitor in Command Line to print anomaly detection results



Lab 6: Prat 1 – Hardware Lab: Setting up Arduino CLI

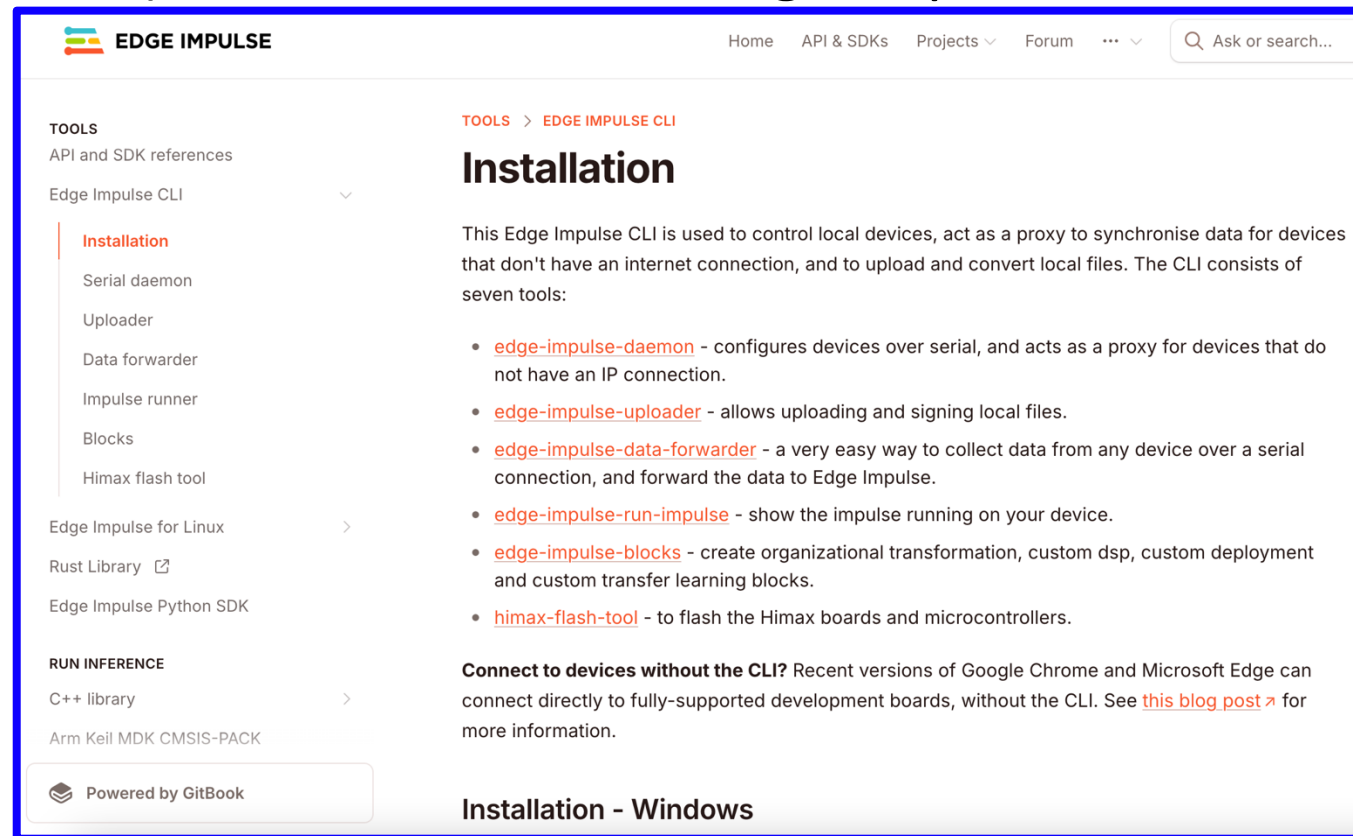
- Go to <https://docs.arduino.cc/arduino-cli/installation/>
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Install Arduino CLI



Skip this Step if you have already done this in Lab 3!

Lab 6: Prat 1 – Hardware Lab: Setting up Edge Impulse CLI

- Go to <https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Edge Impulse CLI

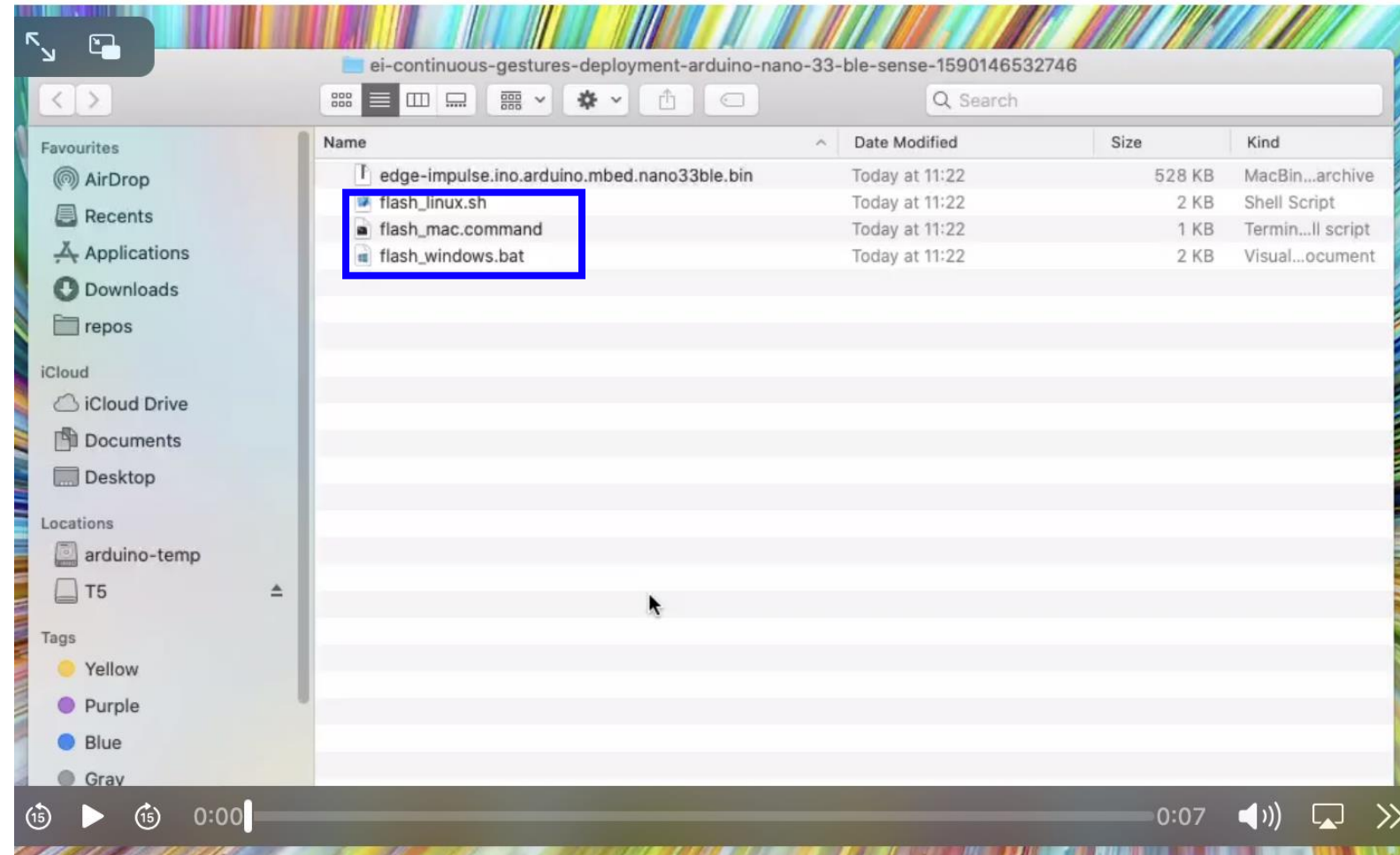


The screenshot shows the Edge Impulse CLI installation page. The header includes the Edge Impulse logo and navigation links: Home, API & SDKs, Projects, Forum, and a search bar. The left sidebar lists various tools and resources, with 'Edge Impulse CLI' expanded to show 'Installation', 'Serial daemon', 'Uploader', 'Data forwarder', 'Impulse runner', 'Blocks', and 'Himax flash tool'. The main content area is titled 'Installation' and explains that the CLI is used to control local devices, act as a proxy to synchronise data, and upload and convert local files. It lists seven tools: `edge-impulse-daemon`, `edge-impulse-uploader`, `edge-impulse-data-forwarder`, `edge-impulse-run-impulse`, `edge-impulse-blocks`, and `himax-flash-tool`. A section titled 'Connect to devices without the CLI?' mentions that recent versions of Google Chrome and Microsoft Edge can connect directly to development boards. The footer indicates the page is 'Powered by GitBook'.

Skip this Step if you have already done this in Lab 3!

Lab 6 : Prat 1 – Hardware Lab: Flashing Model to Arduino Nano BLE

- Flash the compressed model (downloaded from Edge Impulse) onto the board by double clicking on the script for your operating system
- Follow the instructions prompted in command line to observe the results in command line (*you may need to type the prompted commands in a new command prompt window*)



Lab 6: Prat 2 - Software - Road Map

1. Load data and visualize using dimensionality reduction techniques (PCA, tsne, UMAP)
2. Train autoencoder using quantization aware training
3. Perform int-8 quantization
4. Generate model bytes



Lab 6: Prat 2 – Software Lab – Edge Impulse

- Open TinyML_Lab6_Student_TODO.ipynb in tinymml-arduino python environment
- Use the provided dataset - mHealth_subject6.log
- Read instructions and comments carefully to complete all the TODO tasks (5 tasks)
- Run the cells to perform int 8 quantization aware training on autoencoder
- Generate `autoencoder_model.cc` that contains the int 8 model bytes

Lab 6: Prat 2 - Hardware - Road Map

1. Download “Activity_Anomaly.ino”
2. Find the “autoencoder_model.cc” generated from your completed TinyML_Lab6_Student_TODO.ipynb
3. Put “Activity_Anomaly.ino” and “autoencoder_model.cc” in a one folder named “Activity_Anomaly”
4. Double Click on “Activity_Anomaly.ino” and deploy model on board observe serial monitor results

